

SensiSpace

Project Development Timeline

Satellite Land Cover Classification using Deep Learning

March 12, 2026 - May 09, 2026 (58 days)

Phase 1	Research & Design	Mar 12 - Mar 27 (16d)
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Phase 2	Development & Deployment	Mar 28 - Apr 23 (27d)
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Phase 3	Testing & Documentation	Apr 24 - May 09 (16d)
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Key Milestones

Mar 27	M1: Architecture Design Complete
Apr 08	M2: Models Trained & Validated
Apr 23	M3: Web Application Deployed
May 09	M4: Final Report & Demo Ready

Gantt Chart



Milestones & Deliverables

- Mar 27

M1: Architecture Design Complete - BDH architecture finalized, dataset ready
- Apr 10

M2: Models Trained - ResNet: 81.98%, BDH: 95.87% accuracy achieved
- Apr 23

M3: Web App Deployed - Full-stack application with spectral visualization
- May 09

M4: Project Complete - Final report, demo script, all documentation

Phase 1: Research & Design

March 12 - March 27, 2026 | 16 days

Duration	Task	Deliverable
Mar 12 - 17	Literature & reference survey	Research notes
	CNN architectures (ResNet, VGG, Inception)	
	Attention mechanisms (SE-Net, CBAM)	
	Lateral inhibition in neuroscience	
	Remote sensing classification papers	
Mar 14 - 18	EuroSAT dataset analysis	Dataset report
	27,000 images, 10 classes, Sentinel-2	
Mar 17 - 23	BDH architecture design	Architecture diagram
	Multi-Scale Feature Pyramid (3x3/5x5/7x7)	
	Lateral Inhibition Block	
	Contextual Attention (channel + spatial)	
	Recurrent Refinement (gated feedback)	
Mar 19 - 21	Environment & dependency setup	Dev environment
Mar 22 - 26	Data pipeline & augmentation	Data loading code
	80/20 split, flips, rotation, color jitter	
Mar 25 - 27	Architecture review & validation	Validated design

Milestone M1 (Mar 27): Architecture finalized, dataset preprocessed, dev environment ready

Phase 2: Development & Deployment

March 28 - April 23, 2026 | 27 days

Duration	Task	Deliverable
Mar 28 - Apr 1	ResNet-18 baseline implementation	resnet.py
	Pretrained ImageNet, FC: 512 -> 10	
Mar 30 - Apr 6	BDH architecture coding	baby_dragon.py
	MultiScaleBlock, LateralInhibition	
	ContextualAttention, RecurrentRefinement	
Apr 2 - 5	Training pipeline development	train.py
	Adam, LR=0.001, StepLR, CrossEntropy	
Apr 5 - 10	Model training (15 epochs each)	Trained weights
	ResNet-18: 81.98% accuracy	
	BDH: 95.87% accuracy	
Apr 9 - 12	Hyperparameter tuning	Optimized config
Apr 11 - 15	Spectral segmentation module	gradcam.py
	HSV/LAB analysis, Green Leaf Index	
Apr 14 - 17	Backend API development	app.py
	/predict, /sample, /metrics, /colors	
Apr 16 - 20	Frontend UI development	HTML/CSS/JS
	3-tab dark-mode interface, charts	
Apr 20 - 22	Integration & end-to-end testing	Working app
Apr 22 - 23	Deployment & verification	Deployed system

Milestone M2 (Apr 10): Both models trained - BDH: 95.87%, ResNet: 81.98%

Milestone M3 (Apr 23): Full-stack web application deployed with spectral visualization

Phase 3: Testing, Optimization & Documentation

April 24 - May 09, 2026 | 16 days

Duration	Task	Deliverable
Apr 24 - 28	Performance optimization	Optimized models
	Inference speed benchmarking	
	Memory usage optimization	
Apr 27 - May 1	Extended testing & validation	Test report
	Edge case testing	
	Cross-browser compatibility	
Apr 30 - May 5	Documentation & final report	Project report PDF
	Error handling verification	
	Architecture explanation document	
May 4 - 7	Demo preparation & rehearsal	Demo script PDF
	Training results & analysis	
	System architecture documentation	
May 7 - 9	Final review & submission	Final deliverables
	Presentation flow & talking points	
	Q&A preparation	
	Code review & cleanup	
	Final project submission	

Milestone M4 (May 09): Final report submitted, demo presentation complete

Milestones & Deliverables Summary

M1	Architecture Design Complete - BDIdSACh dataset (analysis) designed & validated - Data augmentation pipeline fully implemented	March 27, 2026
M2	Models Trained & Validated - BabyDrug classification @ 96.6% accuracy - Spectral segmentation metrics generated	April 10, 2026
M3	Web Application Deployed - PastAPI backend & frontend (APIs) - Collected & deployed distribution charts	April 23, 2026
M4	Project Complete - Comprehensive project report (PDF) marked - End script with Q&A prepared	May 09, 2026